Payload Limit of a Model-Scale Semi-Submersible: Stability Analysis, Tank Test and Correction

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Based on a five-member team project in the Ocean Structure Design course (report submitted 24 November 2025). The author co-led the configuration design and hydrostatic analysis with one teammate and carried out the re-analysis presented in Sections 5 and 6 in November 2026.

Abstract. A six-column semi-submersible model built from PVC pipe was designed to carry the largest possible deck payload, applied in units of 0.18 kg (food units used as standard weights). The submitted analysis estimated the payload from reserve buoyancy at 6.1 units and found the metacentric height positive up to 6 units, but in the tank the model capsized between the second and third unit. Re-examination showed that the displaced volume used for the centre of buoyancy and the metacentric radius, 1.17 L, violated hydrostatic equilibrium, which requires 2.15 L for the 2.15 kg model. Because the waterplane of a semi-submersible with submerged pontoons consists only of its columns, this error overstated the metacentric radius by a factor of 1.83 and the metacentric height by a factor of about three. With the displaced volume corrected, the pitch metacentric height reaches zero at 2.9 units, consistent with the test. The case shows that the payload of a semi-submersible is limited by stability rather than by buoyancy.

Keywords: semi-submersible; intact stability; metacentric height; model test; hydrostatics

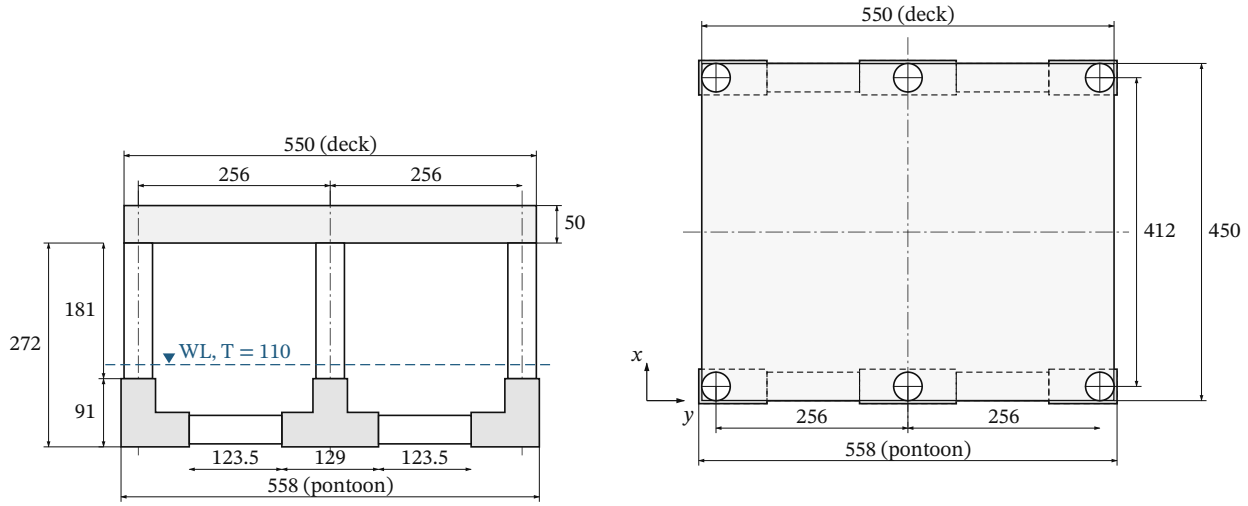
1. Introduction

A semi-submersible supports its deck on a small number of columns standing on submerged pontoons. Because the pontoons lie below the waterline, the waterplane is formed only by the column sections, which keeps the heave response to waves low but also makes the waterplane moment of inertia, and therefore the metacentric radius, small. Payload placed on the deck raises the centre of gravity at the same time. The usable payload is therefore bounded both by reserve buoyancy, the volume of column still above water, and by stability.

In the Ocean Structure Design course, teams designed and built a model semi-submersible from PVC pipe and competed on the number of 0.18 kg payload units each model could carry. This report describes the design of our model, compares the submitted hydrostatic analysis with the tank test, identifies the error that explains the difference, and presents the corrected analysis. Section 2 describes the model, Section 3 the submitted analysis and Section 4 the test result. Sections 5 and 6 give the cause of the discrepancy and the corrected results, and Section 7 discusses what the comparison shows.

2. Model configuration

The model consists of six columns arranged in two rows of three on a 450 mm × 550 mm deck (Fig. 1). Each row of columns is joined at the bottom by a pontoon made of T and 90° pipe fittings and short pipe sections. Using fittings rather than end caps lowers the centre of gravity and adds submerged volume, which increases the air gap available for payload. The principal particulars are listed in Table 1.



Dimensions in mm. Deck: XPS foam board; columns (OD 38) and pontoons: PVC pipe and fittings. Waterline for the unloaded model. The pontoon fittings (OD 46) are wider than the column pipe and protrude 4 beyond the deck at each end. Hidden lines dashed.

Fig. 1. General arrangement of the model: front elevation (left) and plan (right). Drawn from the team's design sketch.

Table 1. Principal particulars of the model.

Item	Value
Deck	450 mm × 550 mm
Number of columns	6
Column waterplane area, A (each)	1,134 mm²
Column outer diameter, length	38 mm, 181 mm
Pontoon height, H_p	91 mm
Pontoon displaced volume, V_p	2.02 L
Model mass, m_0	2.151 kg
Centre of gravity above keel, KG_0	125.5 mm
Payload unit: mass, height of centre of gravity	0.18 kg, 341.5 mm

The deck proportions were chosen for stability. With two of the six columns lying on the x-axis, only four columns contribute to the roll moment of inertia of the waterplane through the parallel-axis term, while all six contribute in pitch:

$$I_{xx} = 6I_{gg} + 4A\,x^2, \qquad I_{yy} = 6I_{gg} + 6A\,y^2, \tag{1}$$

where I_{gg} is the second moment of a single column about its own centroid and x and y are the offsets of the columns from the respective axes. On a square 500 mm × 500 mm deck (x = y = 231 mm), I_{xx} is therefore much smaller than I_{yy}. Lengthening the deck to 550 mm in y and shortening it to 450 mm in x moved the off-axis columns away from the x-axis (256 mm) and the column rows towards the y-axis (206 mm), so that $I_{xx} = 2.98\times10^{-4}$ m⁴ and $I_{yy} = 2.89\times10^{-4}$ m⁴ became nearly equal.

3. Submitted analysis

The submitted analysis first computed the draft. The pontoons provide 2.02 kg of buoyancy when fully submerged, so the six columns carry the remaining 0.13 kg and are immersed by 19 mm. The draft is 110 mm and the air gap, the dry length of column, is 162 mm. Dividing the reserve buoyancy of the dry column length by the mass of one unit gave an estimated payload of 6.13 units.

The stability for each payload was then assessed from the metacentric height

$$GM = KB + BM - KG, \qquad BM = \frac{I}{V}, \tag{2}$$

with the centre of gravity of the loaded model

$$KG = \frac{m_0\, KG_0 + n\, m_c\, z_c}{m_0 + n\, m_c}, \tag{3}$$

where n is the number of payload units, m_c = 0.18 kg and z_c = 341.5 mm. The submitted values of GM remained positive for all payloads up to six units, decreasing from 169 mm (roll) and 161 mm (pitch) with no payload to 14 mm and 10 mm with six units (Fig. 2). The analysis therefore predicted that the model would carry six units.

4. Tank test

In the tank the model capsized between the second and the third unit. The measured limit was less than half of the predicted one. Since the air gap with three units is still more than 80 mm, the failure was not caused by loss of buoyancy, which pointed to an error in the stability calculation rather than in the payload estimate.

5. Cause of the discrepancy

In equilibrium the displaced volume is fixed by the mass of the floating body alone:

$$\rho V = m \quad \Rightarrow \quad V = \frac{m}{\rho} = \frac{2.151\ \mathrm{kg}}{1000\ \mathrm{kg/m^3}} = 2.151\ \mathrm{L}. \tag{4}$$

The draft calculation in Section 3 satisfied this condition. The calculation of KB and BM, however, used a total submerged volume of 1.174 L, 0.98 L less than the equilibrium value. The same gap of 0.98 L appears for every payload case. The most likely explanation is that the volume of the pipe fittings, which was included in the buoyancy table used for the draft, was left out of the separate table used for KB. The two tables computed the same quantity independently and were never checked against each other or against Eq. (4).

The effect of this error is large because of the small waterplane. With the pontoons submerged, BM = I/V is the only term that resists the rise of KG, and dividing I_{xx} by 1.174 L instead of 2.151 L raises BM in roll from 138 mm to 254 mm. The overstated BM kept GM positive for every payload, so the result appeared reasonable and was not questioned.

6. Corrected analysis

For each payload the displaced volume was set to V = (m_0 + n m_c)/ρ. The pontoons remain fully submerged, so the additional volume is taken by the columns, whose immersion is

$$h_c = \frac{V - V_p}{6A}, \tag{5}$$

and the centre of buoyancy is

$$KB = \frac{V_p\, z_p + 6A\, h_c\, (H_p + h_c/2)}{V}, \tag{6}$$

where z_p = 33.3 mm is the volume-weighted centroid of the pontoons. Since the original spreadsheet was no longer available, the missing 0.98 L was assigned to the pontoons at this centroid. KG follows from Eq. (3) and is unchanged by the correction. Table 2 compares the submitted and corrected values with no payload.

Table 2. Hydrostatic quantities with no payload, submitted and corrected.

Quantity	Submitted	Corrected
Displaced volume, V	1.174 L	2.151 L
KB	40.6 mm	37.3 mm
BM, roll	253.7 mm	138.4 mm
BM, pitch	246.4 mm	134.5 mm
KG	125.5 mm	125.5 mm
GM, roll	168.8 mm	50.2 mm
GM, pitch	161.5 mm	46.2 mm

Fig. 2 shows GM as a function of payload. In the corrected analysis GM in pitch falls to zero at 2.9 units and GM in roll at 3.2 units, so pitch governs. This is consistent with the observed capsize between the second and third unit. The model capsized slightly before the predicted limit, which is expected because GM = 0 marks the loss of initial stability for an ideally centred load, while in the test the units were placed by hand and any offset heels the model earlier.

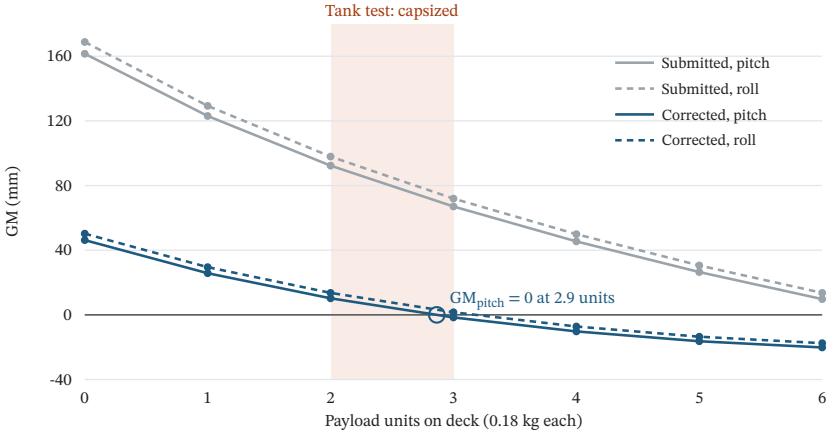


Fig. 2. Metacentric height versus payload, submitted and corrected. The shaded band marks the capsize observed in the tank test.

The correction also strengthens the reason for the deck layout chosen in Section 2. Applying the corrected volume to the square 500 mm × 500 mm deck gives GM in roll of only 25 mm with no payload, against 50 mm for the 450 mm × 550 mm deck, assuming the same KG. The square deck would have lost roll stability between the first and second unit.

The natural period in heave, 1.48 s with an added-mass coefficient of 0.7, was computed in the original report with the correct immersed volume and is not affected.

7. Discussion

The re-analysis rests on one assumption: the location of the missing 0.98 L. Placing all of it in the upper part of the pontoons, which is an upper bound, raises GM in roll with no payload from 50 mm to 66 mm and moves the pitch stability limit from 2.9 to about 4.2 units. The limit then remains well below the buoyancy estimate of 6.1 units, and the centroid assumption agrees more closely with the test, so the conclusion does not depend on this choice. The payload units were modelled as point masses on the centreline, and free-surface effects and the elasticity of the joints were not considered.

The comparison carries a general lesson for the design of semi-submersibles. The payload estimate based on reserve buoyancy (6.1 units) was more than twice the stability limit (2.9 units). For a hull whose waterplane is formed by a few columns, the restoring moment is small and decreases as displacement grows, while deck payload raises the centre of gravity; stability, not buoyancy, sets the payload limit. A

simple check of the displaced volume against the mass, Eq. (4), would have revealed the error before the test.

8. Conclusions

The submitted analysis of a six-column model semi-submersible predicted a payload of six units, whereas the model capsized between the second and the third. The discrepancy was traced to a displaced volume of 1.17 L in the calculation of KB and BM, which violated the equilibrium value of 2.15 L and overstated GM by a factor of about three. The corrected analysis predicts loss of initial stability at 2.9 units, consistent with the test, and confirms that the deck layout chosen to balance roll and pitch stability was necessary. For semi-submersibles the payload limit should be set from GM, and every hydrostatic table should be checked against $V = m/\rho$.

Acknowledgements

The model was designed and built by a five-member team in the Ocean Structure Design course at Korea Maritime & Ocean University. The author thanks the team members for their work on the model and the original report.

Appendix

The original course report (in Korean, submitted 24 November 2025) and the spreadsheet used for the re-analysis, in which the number of payload units is an input, are available from the author.